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APPARATUS FOR TREATING CONTAINERS ADVANCING ALONG A LINEAR DIRECTION

Abstract

An apparatus (1) for treating containers (100) advancing along a linear direction (A-A), the apparatus (1) comprising: a first linear guide (2) developing parallel to the linear direction (A-A) and being fixedly located in a first environment (30) that is a controlled-contamination environment; a carriage (3) rotatably coupled to said first linear guide (2); a first supporting element (4) for supporting one container (100) by its neck (100a), the first supporting element (4) being integrally mounted on the carriage (3); a second linear guide (6) developing parallel to said linear direction (A-A) and being fixedly located in a second environment (40) separated from the first environment (30); a slider (7) slidably coupled to the second linear guide (6), the slider (7) and the carriage (3) being magnetically or electromagnetically coupled.

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Background/Summary

[0001] The present invention relates to an apparatus for treating containers advancing along a linear direction.

[0002] The proposed invention finds application in the pharmaceutical industry, in particular for transporting vials within a clean room.

[0003] Another field of application is the food industry, in particular the bottling sector and, more specifically, bottling using aseptic technology where containers are moved within a clean room.

[0004] For example, a reference sector for the present invention is the bottling of so-called “sensitive” food products, i.e. products that are particularly sensitive to bacteriological contamination and oxidation, such as, for example, isotonic drinks, juices, nectars, soft drinks, tea, milk-based beverages, coffee-based beverages, etc., for which avoiding possible microbiological contamination throughout all packaging steps is of fundamental importance.

[0005] The present invention relates in particular to bottling lines with aseptic technology where maintaining sterility is of fundamental importance.

[0006] However, the present invention can also be used in non-aseptic filling and capping systems.

[0007] In general, the main issue in moving objects within a clean room is linked to the interfaces of the clean room with the outer environment for allowing the passages of the drive systems and at the same time preventing contaminants from entering the clean room.

[0008] Another goal is maintaining the overall volume of the clean room as small as possible.

[0009] From document DE 102014102630 A1 it is known a drive system arranged outside a clean room and designed to move an inner transport element located within the clean room by means of a magnetic or electromagnetic interaction. Both the drive system and the inner transport element comprise carriages mounted on corresponding rails which are mutually parallel and in contact with a separation vertical wall delimiting the clean room.

[0010] Another solution, disclosed in document U.S. Pat. No. 11,498,781, proposes a transport device located in the clean room and magnetically coupled to a drive element designed as an autonomous vehicle which is located outside the clean room.

[0011] The known solutions are either not versatile or too complex.

[0012] In this context, the technical task underpinning the present invention is to provide an apparatus for treating containers advancing along a linear direction, which obviates the drawbacks of the prior art mentioned above.

[0013] In particular, it is an object of the present invention to propose an apparatus for treating containers advancing along a linear direction, the apparatus being at the same time versatile, reliable, and simple.

[0014] The defined technical task and the specified aims are substantially achieved by an apparatus for treating containers advancing along a linear direction, the apparatus comprising: [0015] a first linear guide developing parallel to the linear direction and being fixedly located in a first environment that is a controlled-contamination environment; [0016] a carriage rotatably coupled to the first linear guide; [0017] a first supporting element for supporting one container by its neck, said first supporting element being integrally mounted on the carriage; [0018] a second linear guide developing parallel to the linear direction and being fixedly located in a second environment; [0019] a slider slidably coupled to the second linear guide, the slider and the carriage being

magnetically or electromagnetically coupled; [0020] at least one separation wall for separating the first environment from the second environment, said first linear guide being arranged at a first predefined distance from separation wall.

[0021] According to one embodiment of the invention, the separation wall is located at a predefined height from a ground level. In particular, the second linear guide is arranged below the separation wall and the first linear guide is arranged above the separation wall or vice versa.

[0022] Preferably, the separation wall has at least one inclined portion.

[0023] According to a preferred embodiment, the apparatus further comprises a second supporting element for supporting one container by its neck. The second supporting element is integrally mounted on the carriage.

[0024] Preferably, the carriage comprises: [0025] a framework having a stand developing towards the separation wall; [0026] a plurality of wheels rotatably mounted on the framework; [0027] a case supported by the stand.

[0028] In particular, the case is magnetically or electromagnetically coupled to the slider.

[0029] According to one aspect of the invention, the case and the slider do not contact the separation wall.

[0030] In particular, the case is a plate-like body arranged in the first environment so as to face an inner surface of the separation wall and the slider is a plate-like body arranged in the second environment so as to face an external surface of the separation wall.

[0031] Preferably, the first linear guide comprises a first rail and a second rail which are mutually parallel.

[0032] The plurality of wheels comprises a first pair of wheels rotatably coupled to the first rail and a second pair of wheels rotatably coupled to the second rail.

[0033] In one embodiment, each wheel of said first pair of wheels is a double flange wheel having a groove. The first rail comprises an edge developing parallel to the linear direction from which two inclined surfaces originate. The edge of the first rail fits in the grooves of the corresponding wheels of the first pair of wheels.

[0034] The second rail has a flat upper surface, each wheel of the second pair of wheels being configured to roll on the flat upper surface.

[0035] In another embodiment, each wheel of said second pair of wheels is also a double flange wheel having a groove. The second rail comprises an edge developing parallel to the linear direction from which two inclined surfaces originate. The edge of the second rail fits in the grooves of the corresponding wheels of the second pair of wheels.

[0036] Preferably, the apparatus further comprises at least one interlock element fixedly mounted on the carriage and configured to partially enwrap one of the rails.

[0037] In one embodiment, the case comprises at least a first magnet and the slider comprises at least a second magnet.

[0038] In another embodiment, the slider comprises a ferromagnetic material and the case comprises at least one magnet.

[0039] In another embodiment, the case comprises a ferromagnetic material and the slider comprises at least one magnet.

[0040] According to one aspect of the invention, the apparatus further comprises: [0041] a box delimiting the first environment, the separation wall being a bottom wall of said box; [0042] a sterilizing unit configured to sterilize the containers; [0043] a rinsing unit configured to rinse the containers, the rinsing unit being located downstream the sterilizing unit; [0044] a filling unit configured to fill the containers with a fluid product, the filling unit being located downstream the rinsing unit; [0045] a closing unit configured to close the filled containers, the closing unit being located downstream the filling unit.

Description

[0046] Further features and advantages of the present invention will more fully emerge from the non-limiting description of a preferred but not exclusive embodiment of an apparatus for treating containers advancing along a linear direction, as illustrated in the accompanying drawings, in which:

[0047] FIG. 1 illustrates a side view of an apparatus for treating containers advancing along a linear direction, according to the present invention;

[0048] FIG. 2 illustrates an enlarged portion of part of the apparatus of FIG. 1;

[0049] FIG. 3 illustrates a frontal perspective view of a pair of containers advancing in the apparatus of FIG. 1;

[0050] FIG. 4 illustrates a back perspective view of the pair of containers of FIG. 3;

[0051] FIG. 5 illustrates one embodiment of the rails of the first linear guide of the apparatus of FIG. 1, in a cross-sectional view;

[0052] FIG. 6 illustrates another embodiment of the rails of the first linear guide of the apparatus of FIG. 1, in a cross-sectional view;

[0053] FIG. 7 illustrates an enlarged portion of the slider and the case of the apparatus of FIG. 1, in a cross-sectional view.

[0054] With reference to the figures, number 1 indicates an apparatus for treating containers 100 advancing along a linear direction A-A.

[0055] In this context, the term container 100 indicates either a formed container or a parison made of thermoplastic material.

[0056] The apparatus 1 comprises a first linear guide 2 developing parallel to the linear direction A-A, and a carriage 3 rotatably coupled to the first linear guide 2.

[0057] The first linear guide 2 is fixedly located in a first environment 30 that is a controlled-contamination environment.

[0058] Therefore, also the carriage 3 is arranged in the first environment 30.

[0059] The apparatus 1 further comprises a first supporting element 4 for supporting one container 100 by its neck 100a. The first supporting element 4 is integrally mounted on the carriage 3 and is arranged in the first environment 30.

[0060] Preferably, the first supporting element 4 comprises a first gripper 5 for gripping the neck 100a of one of the containers 100.

[0061] According to a preferred embodiment, the apparatus 1 further comprises a second supporting element 14 for supporting one container 100 by its neck 100a. Analogously to the first supporting element 14, the second supporting element 14 is also integrally mounted on the carriage 3 and is arranged in the first environment 30.

[0062] Preferably, the second supporting element 14 comprises a second gripper 15 for gripping the neck 100a of one of the containers 100.

[0063] In an alternative embodiment (not illustrated), the first supporting element 4 and the second supporting element 14 comprise corresponding pockets for receiving the neck 100a of the containers 100.

[0064] In one embodiment of the invention, each of the supporting elements 4, 14 comprises a bar 17 supporting the corresponding gripper 5, 15 and a reinforcement structure 18 applied to the bar 17 for reinforcing it.

[0065] Each bar 17 has a main development in a direction which is orthogonal to the linear direction A-A. Each reinforcement structure 18 has a substantially triangular extension beside the corresponding bar 17.

[0066] The apparatus 1 further comprises a second linear guide 6 developing parallel to the linear direction A-A, and a slider 7 slidably coupled to the second linear guide 6.

[0067] The second linear guide **6** is fixedly located in a second environment **40**, which is separated from the first environment **30**.

[0068] Therefore, the slider **7** is also arranged in the second environment **40**.

[0069] The second linear guide **6** is parallelly arranged with respect to the first linear guide **2**.

[0070] Advantageously, the slider **7** and the carriage **3** are magnetically or electromagnetically coupled.

[0071] Even if magnetically coupled, the slider **7** and the carriage **3** are physically separated (i.e., not in contact) since they lie respectively in the second environment **40** and in the first environment **30**.

[0072] The separation between the two environments **30**, **40** is obtained by a separation wall **8** located at a predefined height H from a ground level G .

[0073] According to one embodiment of the invention, the separation wall **8** has at least one inclined portion **81**. In particular, the inclined portion **81** is inclined with respect to the ground level G , which is a floor or a terrain.

[0074] In this context, the first environment **30** is also referred to as “inner environment”, whereas the second environment **40** is also referred to as “outer environment”.

[0075] As said, the inner environment **30** is a controlled-contamination environment.

[0076] In the separation wall **8** two surfaces are identified, that are: [0077] a first surface or inner surface **8a**, which faces the inner environment **30**; and [0078] a second surface or outer surface **8b**, which faces the outer environment **40**.

[0079] The first linear guide **2** and the carriage **3** are located in the inner environment **30**.

[0080] In this context, the first linear guide **2** is also referred to as “inner guide”.

[0081] According to one aspect of the invention, the first linear guide **2** is arranged in the first environment **30** at a first predefined distance $d_{sub.1}$ from the separation wall **8**. In particular, the first linear guide **2** does not contact the separation wall **8**.

[0082] In this context, the second linear guide **6** is also referred to as “outer guide”.

[0083] According to one aspect of the invention, the second linear guide **6** is arranged in the second environment **40** at a second predefined distance $d_{sub.2}$ from the separation wall **8**. In particular, the second linear guide **6** does not contact the separation wall **8**. In one embodiment, the second predefined distance $d_{sub.2}$ is lower than the first predefined distance $d_{sub.1}$.

[0084] In other words, the second linear guide **6** is closer to the separation wall **8** than the first linear guide **2**.

[0085] According to the illustrated embodiment, the separation wall **8** divides the inner environment **30** which is located above the separation wall **8**, from the outer environment **40** which is located below the separation wall **8**.

[0086] In this embodiment, the first linear guide **2** is arranged at a higher height than the predefined height H of the separation wall **8** with respect to the ground level G .

[0087] The second linear guide **6** and the slider **7** are located in the outer environment **40**, thus below the separation wall **8**.

[0088] According to another embodiment (not illustrated), the separation wall **8** divides the inner environment **30** which is located below the separation wall **8**, from the outer environment **40** which is located above the separation wall **8**.

[0089] In this embodiment, the second linear guide **6** and the slider **7** are located in the outer environment **40**, thus above the separation wall **8**, whereas the first linear guide **2** is located in the inner environment **30**, thus below the separation wall **8**.

[0090] According to one aspect of the invention, the carriage **3** comprises: [0091] a framework **9**; [0092] a plurality of wheels **10**, **11** rotatably mounted on the framework **9**; [0093] a case **12** supported by the framework **9** and magnetically coupled to the slider **7**.

[0094] Preferably, the framework **9** has a substantially T-shape in cross-section.

[0095] In particular, the framework **9** comprises a chassis **13** and a stand **16** originating from the

chassis **13** and developing towards the separation wall **8**.

[0096] Preferably, the supporting elements **4**, **14** are mounted on the chassis **13** at an opposite side with respect to the stand **16**.

[0097] In particular, the bars **17** and the reinforcement structures **18** are mounted on the chassis **13** at an opposite side with respect to the stand **16**.

[0098] The chassis **13** has an extension that is substantially parallel to the ground level G.

Assuming the ground level G as a horizontal plane, the chassis **13** has also a horizontal extension.

[0099] In the illustrated embodiment, the chassis **13** comprises a tubular structure of four tubular pieces **13a**, **13b**, **13c**, **13d**. In particular, a first and a second tubular pieces **13a**, **13b** are mutually parallel and they are also parallel to the linear direction A-A. A third and a fourth tubular pieces **13c**, **13d** are mutually parallel and they are orthogonal to the linear direction A-A.

[0100] The third and fourth tubular pieces **13c**, **13d** are arranged to connect to first tubular piece **13a** to the second tubular piece **13b**.

[0101] The stand **16** is arranged to connect the chassis **13** to the case **12**.

[0102] In the illustrated embodiment, the stand **16** comprises two further tubular pieces, identified here as a fifth tubular piece **16a** and a sixth tubular piece **16b**.

[0103] The fifth tubular piece **16a** has one end connected to the third tubular piece **13c** and another end connected to the case **12**.

[0104] The sixth tubular piece **16b** has one end connected to the fourth tubular piece **13d** and another end connected to the case **12**.

[0105] The case **12** and the slider **7** are separated by the separation wall **8**.

[0106] Preferably, both the case **12** and the slider **7** do not contact the separation wall **8**.

[0107] The case **12** is preferably a plate-like body arranged in the first environment **30** so as to face the inner surface **8a** of the separation wall **8**.

[0108] In other words, the case **12** is supported by the stand **16** in a position which is suspended above the separation wall **8**.

[0109] The slider **7** is preferably a plate-like body arranged in the second environment **40** so as to face the outer surface **8b** of the separation wall **8**.

[0110] According to one embodiment of the invention, the case **12** comprises at least a first magnet **12a** and the slider **7** comprises at least a second magnet **7a**.

[0111] Preferably, both the case **12** and the slider **7** comprises a plurality of magnets having alternate polarities.

[0112] According to one aspect of the invention, the case **12** is tight-sealed.

[0113] According to another embodiment of the invention, the slider **7** comprises a magnet and the case **12** comprises a ferromagnetic material.

[0114] According to another embodiment of the invention, the case **12** comprises a magnet and the slider **7** comprises a ferromagnetic material. As said, the carriage **3** is rotatably coupled to the first linear guide **2**.

[0115] The first linear guide **2** comprises a first rail **20** and a second rail **21** that are mutually parallel. These rails **20**, **21** are also parallel to the linear direction A-A.

[0116] The wheels **10**, **11** of the carriage **3** are rotatably coupled to the rails **20**, **21**.

[0117] In particular, a first pair of wheels **10** is rotatably coupled to the first rail **20**, and a second pair of wheels **11** is rotatably coupled to the second rail **21**.

[0118] According to an embodiment of the invention, illustrated in FIG. 5, the first rail **20** comprises an edge **20a** developing parallel to the linear direction A-A. From the edge **20a** originate two inclined surfaces **20b**, **20c** of the first rail **20**.

[0119] The edge **20a** is arranged towards the carriage **3**, in particular towards the first pair of wheels **10**.

[0120] Each wheel of the first pair of wheels **10** is a double flange wheel having a groove **10a** that is shaped so as fit on the edge **20a** of the first rail **20**. The double flange wheels **10** contact the

inclined surfaces **20c**, **20c** while rolling along the first rail **20**.

[0121] The second rail **21** has a flat upper surface **21a** which faces the carriage **3**, in particular the second pair of wheels **11**.

[0122] Each wheel of the second pair of wheels **11** is configured to roll on the flat upper surface **21a**.

[0123] According to one aspect of the invention, the first rail **20** and the second rail **21** are identical but they are differently oriented in the first environment **30**.

[0124] In particular, the two rails **20**, **21** are tubular elements with a square or rectangular cross-section, and they are oriented with a mutual phase shift of 45° with respect the linear direction A-A.

[0125] The choice of coupling the double flange wheels **10** with the edge **20a** of the first rail **20** serves to define more precisely the advancing movement of the overall carriage **3**.

[0126] According to an alternative embodiment, shown in FIG. **6**, the second rail **21** is identical to the first rail **20**, meaning that it has an edge developing parallel to the linear direction A-A and two inclined surfaces originating from said edge.

[0127] In this alternative embodiment, the wheels of the second pair of wheels **11** are identical to the wheels of the first pair of wheels **10**. In particular, each wheel of the second pair of wheels **11** is a double flange wheel so as to fit on the edge of the second rail **21**.

[0128] In this alternative embodiment, the two rails **20**, **21** are preferably identical and they have the same orientation in the first environment **30**.

[0129] In particular, the second rail **21** comprises an edge **21a** developing parallel to the linear direction A-A. From the edge **21a** originate two inclined surfaces **21b**, **21c** of the second rail **21**.

[0130] The edge **21a** is arranged towards the carriage **3**, in particular towards the second pair of wheels **11**.

[0131] Each wheel of the second pair of wheels **11** is a double flange wheel having a groove **11a** that is shaped so as to fit on the edge **21a** of the second rail **21**.

[0132] The double flange wheels **11** contact the inclined surfaces **21b**, **21c** while rolling along the second rail **21**.

[0133] This alternative embodiment achieves an even more precise advancing movement of the overall carriage **3**.

[0134] In the embodiments described above, the carriage **3** is simply rotatably coupled to the first linear guide **2**. This means that the wheels **10**, **11** lie and roll on the rails **20**, **21**.

[0135] In another embodiment, the carriage **3** is also connected to the first linear guide **2** by at least one interlock element **19**. In particular, the interlock element **19** is a bracket fixedly mounted on the chassis **13** and configured to partially enwrap one of the rails **20**, **21**, for example the first rail **20**.

[0136] Accordingly, the carriage **3** can slide on the inner guide **2** with the bracket **19** also sliding along the first rail **20**. This prevents the carriage **3** from falling down the chassis **13** in case of misalignments.

[0137] In another embodiment, there could be more interlock elements **19** in order to increase the reliability of the connection between the carriage **3** and the inner guide **2**.

[0138] The apparatus **1** comprises holding means **22** for supporting the rails **20**, **21**. In particular, the holding means **22** comprises pillars for supporting the rails **20**, **21**.

[0139] In one embodiment, illustrated in FIG. **1**, the pillars **22** are integrally mounted on a base structure **23** which also supports the outer guide **6**.

[0140] The base structure **23** lies on the ground level G.

[0141] In another embodiment, not illustrated, the pillars **22** are anchored to the ground level G.

[0142] The apparatus **1** further comprises motorized means (not illustrated) that are operatively active on the slider **7** to make it slide on the second linear guide **6**.

[0143] The motorized means are of the known type, for example a pneumatic cylinder or an electric motor.

[0144] The motorized means are configured to drive the slide **7** in a direction which is parallel to

the linear direction A-A, either in one way or on the opposite.

[0145] In the illustrated embodiment, the apparatus **1** comprises a box **50** delimiting the first environment **30**. In particular, the separation wall **8** is a bottom wall of the box **50**.

[0146] In particular, the box **50** consists in an isolation device which is configured to physically separate the inner environment **30** from the outer environment **40**.

[0147] In a preferred embodiment, the apparatus **1** comprises a heating unit configured to heat the containers **100**.

[0148] Downstream the heating unit there is arranged a sterilizing unit configured to sterilize the containers **100**. In particular, the sterilizing unit is configured to sterilize the containers **100** by dispensing hydrogen peroxide on them.

[0149] The heating unit upstream the sterilizing unit aims at avoiding the hydrogen peroxide to condense on the surfaces of the containers **100**.

[0150] Downstream the sterilizing unit, there is a rinsing unit configured to rinse the containers **100** by using sterile air.

[0151] Downstream the rising unit, the containers **100** are filled with a fluid product by a filling unit.

[0152] Preferably, a layer of nitrogen is deposited on top of the level of fluid in the containers **100**. This is obtained by a dosing device configured to dose nitrogen.

[0153] The apparatus **1** further comprises a closing unit configured to apply closures or caps to the filled containers **100**.

[0154] The functioning of the apparatus for treating containers advancing along a linear direction, according to the present invention, is described below.

[0155] The containers **100** are fed by pairs to an inlet of the apparatus **1**. For example, the feeding may be performed manually by a human operator or by a robot.

[0156] It is assumed one pair of containers **100** is fed to the grippers **5**, **15** mounted on the carriage **3**.

[0157] The slider **7** is initially actuated by the motorized means so as to slide on the second linear guide **6** on a forward way, which means that the slider **7** moves away from the inlet of the apparatus **1**.

[0158] Accordingly, the two containers **100** supported by the grippers **5**, **15** advance along the linear direction A-A thanks to the magnetic coupling of the case **12** with the slider **7**.

[0159] Upon advancing along the linear direction A-A in the first environment **30**, the containers **100** are subject to a sequence of treatment—heating, sterilizing, rinsing, filling, dosing of nitrogen, closing.

[0160] In a preferred embodiment, the slider **7** is then actuated on a back way on the second linear guide **6**, which means that the slider **7** moves back to the inlet of the apparatus **1**.

[0161] Thus, the closed containers **100** are discharged at the inlet of the apparatus **1**. The discharge may be performed manually by a human operator or by a robot.

[0162] In another embodiment, the containers **100** are discharged at an outlet of the apparatus **1** the containers **100**, either by a human operator or a robot.

[0163] The slider **7** is then actuated on a back way on the second linear guide **6**, which means that the slider **7** moves back to the inlet of the apparatus **1**.

[0164] Accordingly, the case **12** is moved back towards the inlet of the apparatus **1**, where the grippers **5**, **15** may receive a new pair of containers **100**.

[0165] The characteristics of the apparatus for treating containers advancing along a linear direction, according to the present invention, are clear from the description, as are the advantages.

[0166] In particular, the containers are moved within the controlled-contamination environment thanks to the magnetic coupling of the carriage with the slider located outside the controlled-contamination environment.

[0167] The arrangement of the inner guide with the rails at a predefined distance from the

separation (bottom) wall allows the carriage (and therefore the containers) to move independently from the specific shape and irregularities of the separation wall (or bottom wall).

[0168] The attraction forces between the inner block and the outer slider are counterbalanced by the rails and the relative holding means (pillars).

[0169] The proposed apparatus is versatile since it allows to perform a plurality of treating operations (i.e. heating, sterilizing, rising, filling, etc.) in aseptic conditions.

Claims

1. An apparatus for treating containers advancing along a linear direction, said apparatus comprising: a first linear guide developing parallel to said linear direction and being fixedly located in a first environment that is a controlled-contamination environment; a carriage rotatably coupled to said first linear guide; a first supporting element for supporting one container by its neck, said first supporting element being integrally mounted on the carriage; a second linear guide developing parallel to said linear direction and being fixedly located in a second environment; a slider slidably coupled to the second linear guide, said slider and said carriage being magnetically or electromagnetically coupled; at least one separation wall for separating the first environment from the second environment, said first linear guide being arranged at a first predefined distance from said separation wall.
2. The apparatus according to claim 1, wherein said at least one separation wall is located at a predefined height from a ground level, said second linear guide being arranged below said separation wall and said first linear guide being arranged above said separation wall or vice versa.
3. The apparatus according to claim 1, wherein said separation wall has at least one inclined portion.
4. The apparatus according to claim 1, further comprising a second supporting element for supporting one container by its neck, said second supporting element being integrally mounted on the carriage.
5. The apparatus according to claim 1, wherein said carriage comprises: a framework having a stand developing towards the separation wall; a plurality of wheels rotatably mounted on the framework; a case supported by the stand, said case being magnetically or electromagnetically coupled to said slider.
6. The apparatus according to claim 5, wherein the case and the slider do not contact the separation wall.
7. The apparatus according to claim 5, wherein the case is a plate-like body arranged in the first environment so as to face an inner surface of the separation wall and the slider is a plate-like body arranged in the second environment so as to face an external surface of the separation wall.
8. The apparatus according to claim 5, wherein the first linear guide comprises a first rail and a second rail which are mutually parallel, said plurality of wheels comprising a first pair of wheels rotatably coupled to said first rail and a second pair of wheels rotatably coupled to said second rail.
9. The apparatus according to claim 8, wherein each wheel of said first pair of wheels is a double flange wheel having a groove, said first rail comprising an edge developing parallel to the linear direction from which two inclined surfaces originate, the edge of the first rail fitting in the grooves of the corresponding wheels of the first pair of wheels.
10. The apparatus according to claim 9, wherein the second rail has a flat upper surface, each wheel of the second pair of wheels being configured to roll on the flat upper surface.
11. The apparatus according to claim 9, wherein each wheel of said second pair of wheels is a double flange wheel having a groove, said second rail comprising an edge developing parallel to the linear direction from which two inclined surfaces originate, the edge of the second rail fitting in the grooves of the corresponding wheels of the second pair of wheels.
12. The apparatus according to claim 8, further comprising at least one interlock element fixedly

mounted on the carriage and configured to partially enwrap one of the rails.

13. The apparatus according to claim 5, wherein said case comprises at least a first magnet and said slider comprises at least a second magnet.

14. The apparatus according to claim 5, wherein said slider comprises a ferromagnetic material and said case comprises at least one magnet.

15. The apparatus according to claim 1, further comprising: a box delimiting the first environment, the separation wall being a bottom wall of said box; a sterilizing unit configured to sterilize the containers; a rinsing unit configured to rinse the containers, said rinsing unit being located downstream the sterilizing unit; a filling unit configured to fill the containers with a fluid product, said filling unit being located downstream the rinsing unit; a closing unit configured to close the filled containers, said closing unit being located downstream the filling unit.
